

12.1 Chatbot Applications and Taxonomy

What bots are used for, and the main design families.

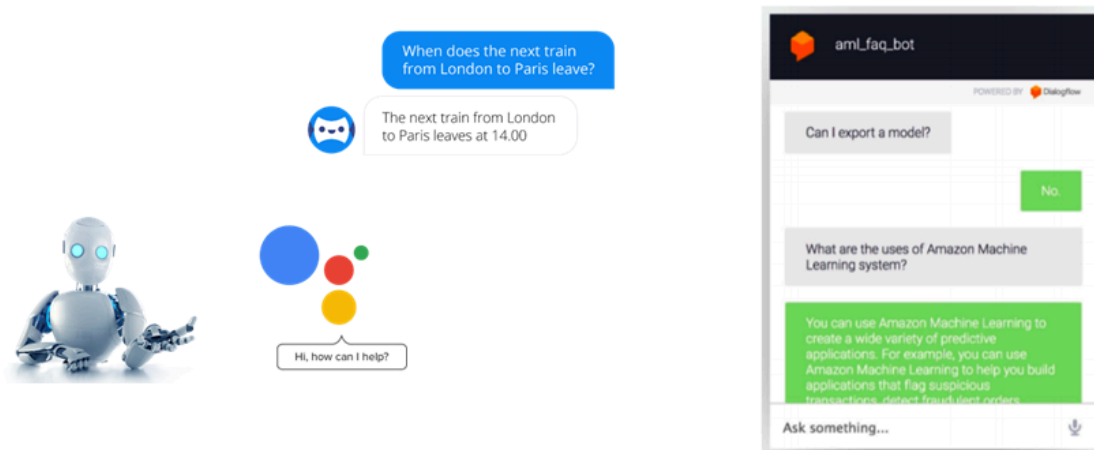
1. A short history

Chatbots and AI grew up together.

1950s–60s. Turing asked whether a machine could talk well enough that a person could not tell. Weizenbaum built **ELIZA** (1966), a rule-and-regex "Rogerian" therapist. People knew it was a program and still treated it as a listener. That is a warning, not a success metric: users will anthropomorphize a shallow bot.

1980s–2000s. Spoken dialog systems appeared, many of them military (DARPA). The new piece was signal processing plus a limited vocabulary, not open chat.

2000s–now. NLP and machine learning made the same ideas usable in customer service, health, finance, and retail. Messenger, Google Assistant, and Alexa are the public face. Most industry bots are still **narrow**.



2. Where they are used

A bot is a UI. The NLP inside it changes with the job.

Domain	What the bot is asked to do
Shopping	Place or change an order, take payment, recommend an item
News	Narrow a search in conversation, return a matching article
Customer service	FAQs, complaints, a scripted flow the business already wrote
Medical	Symptom FAQs; intake questions, especially for older patients
Legal	FAQ, then follow-up questions so the next document is the right one
Education	Tutor, language practice, registration and scheduling

Conversational **recommenders** are a live research and product area. The hard part is not greeting the user. It is tracking constraints as they arrive one turn at a time.

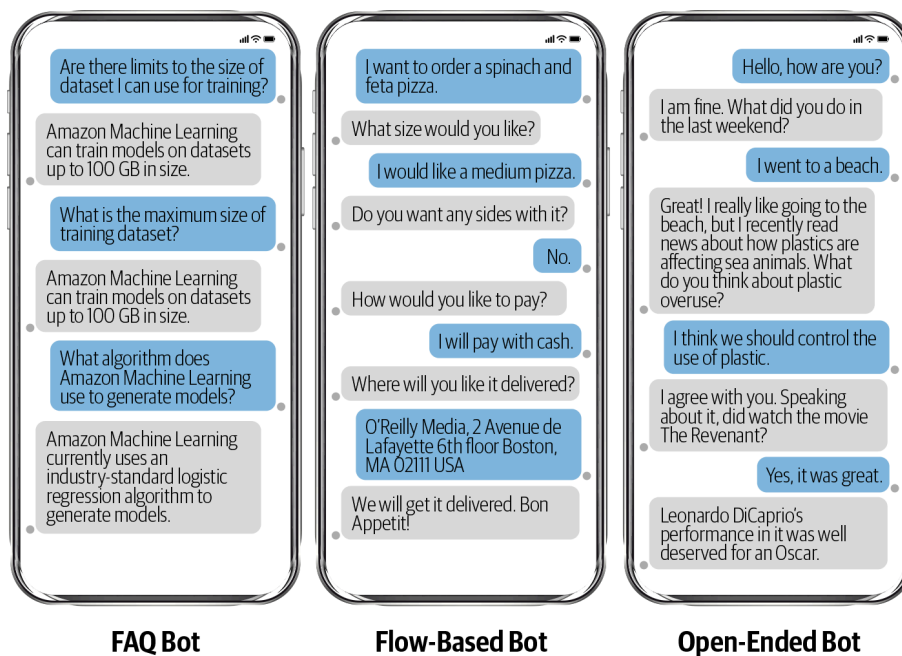
3. Three kinds of bot

How you build the system depends on which of these you are in.

FAQ / exact-answer bot. A fixed set of answers. Understand the question, retrieve the matching response. Turns are mostly independent. This is close to classification plus lookup.

Flow-based bot. The user reveals the request over several turns. Pizza size, then toppings, then address. The bot must **remember** what it already knows and ask only for what is missing. This is a dialog state machine with NLU on each utterance.

Open-ended bot. Chitchat. No ticket to close. The bot does not have to drive toward a goal. Entertainment and brand mascots live here. So do the failure modes: hallucination, inconsistency, and users who think the bot is a person.



4. Goal-oriented vs chitchat

The coarse split used in research and in vendor tools:

Family	Includes	Success looks like
Goal-oriented	FAQ bots, flow-based bots	The user completes a task
Chitchat	Open-ended bots	The conversation stays coherent and on-tone

Both are used in industry. Both are active research. Do not use a chitchat model as a customer-service bot and hope it will "figure out" the order. And do not use an FAQ lookup table when the user needs a five-slot pizza flow. Write the success test first (ticket closed, order placed, user still chatting). The test picks the cell.

A hybrid product is normal: FAQ for "where is my order?", flow for "I want to return this," and a small chitchat layer for "thanks" and "hello." Keep the layers separate. One model that tries to do all three will be good at none of them.

Note **12.2** is the shared pipeline (speech, NLU, dialog manager, NLG). Note **12.3** is intents, slots, and response generation.

5. What this week is for

By the end of note **12.1** you should be able to:

- place ELIZA, spoken dialog, and a modern assistant on the history line
- pick a domain row from the table and say whether the bot is FAQ, flow, or open
- refuse a design that uses chitchat weights for a goal-oriented task

6. Practice

1. Is a password-reset bot FAQ, flow-based, or open-ended? What would make you change that label?
2. Why did ELIZA convince people even though it had no model of the user?
3. A retailer wants "something like Alexa" for returns. Which taxonomy cell do you start in, and what do you refuse to promise?

"Like Alexa" is a channel, not a taxonomy cell. Ask what task must complete.